

BMJ Open Effectiveness of methadone versus buprenorphine in the treatment of opioid use disorder: secondary analyses of prospective cohort study data

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ABSTRACT

Objectives To compare the effectiveness of buprenorphine-naloxone (bup/nal) and methadone maintenance therapy (MMT) in the treatment of patients with opioid use disorder (OUD) during the fentanyl era.

Design Secondary analysis of prospective cohort study data.

Setting Data for the study were collected from 54 clinical sites across Ontario, Canada, between May 2018 and January 2023.

Participants To be included in the present study, participants had to be at least 16 years of age, have provided written informed consent and be receiving either MMT or bup/nal therapy for OUD. This study includes data from 2601 participants, of whom 2068 were receiving MMT and 533 were receiving bup/nal for OUD. The mean age of participants was 39.4 years (SD: 10.9), and 45% were female.

Interventions MMT or bup/nal treatment for OUD.

Outcome measures We employed a propensity score matched analysis to compare treatment outcomes among patients receiving MMT compared with bup/nal. We used ongoing illicit opioid use as an indicator of treatment outcome. We considered participants with >50% of urine drug screens in the past 12 months positive for non-prescribed opioids to be 'non-responders'. We conducted subgroup analyses to identify whether treatment type was associated with ongoing non-prescribed opioid use among patients with and without a history of intravenous drug use (IVDU), and whether treatment type was associated with retention in treatment.

Results Eight per cent of patients on bup/nal were considered non-responders, compared with 11.9% of patients on MMT. We did not find a statistically significant association between treatment type and treatment response. However, we did find that patients on MMT were more likely to stay in treatment for 12 months (OR 1.79, 95% CI 1.45 to 2.22, $p<0.001$). We also found that, among patients without a history of IVDU, those on MMT were more likely to continue using non-prescribed opioids, compared with those on bup/nal (OR 1.72, 95% CI 1.07 to 2.77, $p=0.023$).

Conclusions Among a cohort of patients with OUD receiving treatment during the fentanyl era, we find that there is no statistically significant difference in ongoing non-prescribed opioid use between patients receiving

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ This study's findings are strengthened by the large sample size and long follow-up period (12 months).
- ⇒ We employed objective measures of treatment outcome (urine drug screen data) rather than relying on self-report.
- ⇒ The study employed minimal eligibility criteria whereby participants were not excluded based on cosubstance use or comorbidities, rendering the sample more representative of the target population during the current fentanyl crisis.
- ⇒ This is an observational study and patients were not randomised to either treatment, increasing the risk of potential confounding.
- ⇒ We employed propensity score matching to mitigate bias due to known potential confounders.

MMT compared with bup/nal. Future studies should aim to further compare treatment effectiveness using patient-centred outcomes and pragmatic trial designs.

INTRODUCTION

The prevalence of opioid use disorder (OUD) as well as its detrimental impact on individuals and society has reached an all-time high. In the USA, the opioid crisis has been declared a Public Health Emergency since 2017, and rates of morbidity and mortality have continued to rise since.^{1 2} This is largely due to the increasing prevalence of fentanyl and similar very potent synthetic opioids in the street supply, that substance users are sometimes inadvertently exposed to, leading to unintentional overdose and death.³

Methadone maintenance therapy (MMT) has traditionally been the first-line treatment for OUD. Methadone is a synthetic opioid with a long duration of action which acts as a full agonist on the mu-opioid receptors to mitigate cravings and withdrawal symptoms, while minimising the 'high' associated with short-acting opioids.⁴ In 2002,



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buprenorphine-naloxone (bup/nal) was approved by the US Food and Drug Administration for the treatment of OUD and, in more recent guidelines, has been considered to be first-line treatment for OUD along with MMT.^{5 6} Buprenorphine is a partial opioid agonist with very high affinity for the mu-opioid receptor.⁴ This allows buprenorphine to displace other opioids, while also mitigating cravings and withdrawals through its agonistic properties at the receptor. The high affinity of buprenorphine at the opioid receptor also interferes with the binding of other opioids at the receptor, mitigating euphoria and overdose risk. Being a partial agonist, its effects also plateau at higher levels—mitigating the risk of overdose and sedation that may otherwise occur with high doses of methadone.⁴ The aforementioned properties, among others, make buprenorphine a popular treatment recommendation and potentially safer treatment option for OUD.

The data comparing the effectiveness of MMT to bup/nal for OUD are mixed, however. While many studies have identified that MMT is superior for retention in treatment, a recent systematic review found no difference in outcomes between the two treatments on rates of concurrent substance use, among other outcomes.⁷ A closer look at the individual studies, however, identifies many limitations that restrict the external validity of the findings.^{7–9} Many of these trials have very stringent eligibility criteria, excluding patients with concurrent substance use or comorbidities, for instance, while other trials implement stringent fixed-dose protocols which are not representative of real clinical settings and patients' needs.^{7–9} The varying protocols used may partly explain the mixed outcomes. Moreover, the majority of these studies have reported on retention in treatment as a primary outcome, though prior research suggested that this outcome is of limited importance to the patients seeking treatment.¹⁰ These studies are also outdated as they had been conducted prior to the fentanyl era. Fentanyl is a synthetic opioid that is approximately 100 times more potent than morphine and began dominating the drug supply over the past decade, leading to an exponential rise in deaths due to overdose.¹¹ Given its high potency, individuals are developing very high tolerance levels to opioids, and there is increasing concern that previously recommended treatment regimens and dosing schedules may no longer be as effective.¹² The purpose of our study is to assess the effectiveness of bup/nal compared with MMT in the treatment of patients with OUD using current data, as measured by continued non-prescribed opioid use.

METHODS

Study design

We used data collected from a longitudinal study entitled Pharmacogenetics of Opioid Substitution Treatment Response. This is a prospective cohort study aimed at assessing the association between biopsychosocial

factors and opioid agonist therapy (OAT) outcomes. The protocol for this study has previously been described, though certain modifications to the original protocol have since been made.^{13 14} In brief, data for the study were collected from 54 clinical sites across Ontario, Canada, between May 2018 and January 2023. To be included in the study, participants had to be at least 16 years of age, have provided written informed consent and be receiving either MMT or bup/nal therapy for OUD. We defined OUD as per the Diagnostic and Statistical Manual of Mental Disorders, fifth Edition.¹⁵ All participants underwent a semistructured baseline interview with trained research staff whereby demographic information, as well as past medical and substance use histories, were obtained. At this time, participants also completed the Maudsley Addiction Profile (MAP), a validated tool used to assess treatment outcomes for patients with substance use disorders.¹⁶ As part of the usual treatment for OUD, participants underwent regular urine toxicology screens, typically on a weekly to biweekly basis. The FaStep Assay (Trimedica Supply Network, Concord, Ontario, Canada) was used to detect morphine, oxycodone, fentanyl, methadone metabolite and buprenorphine, as well as other non-opioid substances. Participants were followed at 3 months intervals, for up to 12 months. At each follow-up, the following data were obtained from participants' electronic medical records: current dose, date of last dose taken and results of all urine toxicology screens within the preceding 3 months period.

The current study is reported according to the Strengthening the Reporting of Observational Studies in Epidemiology statement.¹⁷

Statistical methods

All analyses were conducted using STATA V.13.0.¹⁸ Descriptive statistics were used to summarise baseline participant demographics. We used means and SD to express continuous variables that were normally distributed, medians and IQR for skewed data, as well as counts and percentages to summarise categorical variables.

Primary analysis: propensity score matching

We employed a propensity score matched (PSM) analysis to compare treatment outcomes among patients receiving MMT compared with bup/nal. Considering the observational nature of the study, we used a PSM analysis to minimise confounding and account for any potential systemic differences in baseline characteristics between treatment arms. We used ongoing non-prescribed opioid use as an indicator of treatment outcome, as one of the OAT objectives is to reduce non-prescribed opioid use. Previous research has established that concomitant use of opioids is a significant predictor of overdose and mortality.¹⁹ Given this, and consistent with previous research, we considered treatment 'non-responders' as those with >50% of urine drug screens (UDS) in the past 12 months positive for non-prescribed opioids. This cut-off, while arbitrary, we considered clinically important as it suggests high-risk

opioid consumption.^{19–21} Patients receiving MMT were matched in a 1:1 ratio with patients receiving bup/nal using a PSM analysis. The propensity score was generated using the following clinically important factors: age (years), sex, employment status, marital status, history of intravenous drug use in the past 30 days (IVDU; yes/no), concurrent non-prescribed benzodiazepine use (yes/no), cannabis use (yes/no), MAP score for psychological stress (continuous variable) and history of overdose (yes/no).^{22–25} We applied the nearest neighbour matching algorithm with a calliper width of 0.25 of the logit of the propensity score, as is the commonly recommended practice.²⁶ Balance after PSM was assessed by calculating the standardised mean difference (SMD), and we considered balance to have been achieved when SMD is less than 0.1.²⁷ We estimated and reported the average treatment effect on the treated.

Secondary analysis: multivariable logistic regression

We conducted subgroup analyses to identify whether the type of OAT was associated with ongoing non-prescribed opioid use among patients with and without a history of IVDU who remained in treatment for 1 year. Two identical multivariable logistic regression models were employed among patients with and without a history of IVDU, while adjusting the model for the same factors that were used in the PSM, namely: age (years), sex, employment status, marital status, history of opioid overdose (yes/no), concurrent non-prescribed benzodiazepine use (yes/no), cannabis use (yes/no) and MAP score for psychological stress. Another logistic regression was conducted using the entire dataset to assess whether treatment type for OUD is associated with retention in treatment. Once again, treatment retention at 12 months was regressed against treatment type (MMT vs bup/nal) while adjusting for the same clinically important variables.

Patient and public involvement

A patient with lived experiences was called on when designing this research study from inception. They were instrumental in providing feedback on values and patient-important outcomes. They were directly involved in discussions surrounding participant recruitment and selection of outcome measures for the study. They also weighed in on the present study results and their interpretation.

RESULTS

Summary of study participants

This study included data from 2601 participants, of whom 2068 were receiving MMT and 533 were receiving bup/nal for OUD. The mean age of participants was 39.4 years (SD: 10.9), and 45% were female. The median dose of MMT and bup/nal was 70 mg (IQR: 40–100) and 12 mg (IQR: 8–16) per day, respectively. 71% (n=1850) remained in the study for the 12 months of follow-up. Patients in the MMT and bup/nal groups had an average of 50.2 (SD 16.6) and 46.9 (SD 15.6) UDSs collected and analysed during this time, respectively. Over the 1-year period, 13.1% of participants were considered non-responders. Please see [tables 1 and 2](#) for details regarding baseline participant demographics before and after the PSM.

Primary PSM analysis: ongoing non-prescribed opioid use among patients receiving MMT compared with bup/nal

Our PSM analysis included data from 668 participants. Eight per cent (8.1%, 95% CI 5.1 to 11.0) of patients on bup/nal were considered non-responders, compared with 13.5% (95% CI 9.8 to 17.2) of patients on methadone. The χ^2 independence test showed that there was a statistically significant difference in the proportion of non-responders in each treatment arm ($X^2=5.04$, $p=0.025$). However, we did not find a statistically significant association between treatment type and treatment response in our PSM analysis, adjusting for

Table 1 Baseline demographics table (n=2601)

	Methadone (n=2068)	Buprenorphine/naloxone (n=533)	Total sample (n=2601)
	Mean (SD)		
Age (years)	39.6 (10.8)	38.9 (11.1)	39.4 (10.9)
	Median (IQR)		
Average dose (mg/day)	70 (40–100)	12 (8–16)	N/A
MAP psych sx score	11 (4–18)	10 (4–17)	11 (4–18)
	Count (%)		
Female sex	924 (44.7)	245 (46.0)	1169 (44.9)
Currently working	626 (30.3)	213 (40.0)	839 (32.3)
Married or common law	603 (29.2)	161 (30.2)	764 (29.4)
History of IVDU	346 (16.7)	61 (11.4)	407 (15.7)
History of opioid overdose	708 (34.2)	161 (30.2)	869 (33.4)
Cannabis user	1072 (51.8)	262 (49.2)	1334 (51.3)
IVDU, intravenous drug use; MAP, Maudsley Addiction Profile; N/A, not available.			

Table 2 Baseline demographics table of participants included in PSM analysis (n=668)

	Methadone (n=334)	Bup/Nal (n=334)	SMD
	Mean (SD)		
Age (years)	41.7 (10.7)	40.6 (11.2)	0.007
	Median (IQR)		
MAP psych sx score	9.5 (3–16)	9 (3–16)	0.003
	Count (%)		
Female sex	924 (44.7)	245 (46.0)	0.023
Currently working	626 (30.3)	213 (40.0)	0.028
Married or common law	603 (29.2)	161 (30.2)	0.062
History of IVDU	346 (16.7)	61 (11.4)	0.011
History of opioid overdose	708 (34.2)	161 (30.2)	0.048
Cannabis user	1072 (51.8)	350 (65.7)	<0.001

Bup/Nal, buprenorphine-naloxone; IVDU, intravenous drug use; MAP, Maudsley Addiction Profile; PSM, propensity score matched; SMD, standardised mean difference.

age, sex, employment status, marital status, IVDU history, opioid overdose history, non-prescribed benzodiazepine use, cannabis use and MAP score for psychological stress ($p=0.055$).

Secondary analysis: association between treatment type and retention in treatment

Data from 2601 participants were included in this analysis. We found that patients who are on methadone are 1.8 times more likely to stay in treatment at 12 months of follow-up, compared with patients on bup/nal (OR 1.79, 95% CI 1.45 to 2.22, $p<0.001$). We also found that being a woman, being employed, being older and not having a history of IVDU or opioid overdose is associated with a higher likelihood of staying in treatment. Please refer to [table 3](#) below for outcome data.

Secondary analysis: association between treatment type and continued non-prescribed opioid use stratified by history of IVDU

Among patients with a history of IVDU, we found that there is no difference in treatment outcomes by treatment type, after adjusting for clinically important variables ([table 4](#)). Among those without a history of IVDU ($n=1619$), however, we found that patients who are on methadone are 1.7 times more likely to be non-responders, compared with those on bup/nal (OR 1.72, 95% CI 1.07 to 2.77, $p=0.023$). Moreover, among those without a history of IVDU, we found that concurrent non-prescribed benzodiazepine use was associated with a higher likelihood of treatment non-response (OR 2.07, 95% CI 1.19 to 3.59, $p=0.010$).

DISCUSSION

The substance use crisis continues to dominate headlines as opioid overdose remains the leading cause of accidental death in the USA.^{2 28} OATs are the mainstay

of treatment given their established superiority as a harm reduction approach. Novel OATs have been introduced to the market over the years, though little data exist to guide the selection of the optimal OAT based on patient characteristics and circumstances. While many options exist, including slow release oral morphine and injectable diacetylmorphine, MMT and buprenorphine remain

Table 3 Logistic regression: predictors of retention in treatment at 12 months (n=2601)

	Adjusted OR	95% CI	P value
Methadone use*	1.79	1.45 to 2.22	<0.001
Female sex	1.36	1.13 to 1.62	0.001
Cannabis user	0.91	0.76 to 1.08	0.277
Currently working	1.36	1.11 to 1.66	0.003
Age	1.03	1.02 to 1.04	<0.001
IVDU	0.53	0.42 to 0.68	<0.001
Married or common-law	1.09	0.89 to 1.33	0.407
Non-prescribed benzodiazepine use	1.14	0.82 to 1.57	0.438
Opioid overdose history	0.68	0.56 to 0.82	<0.001
MAP psychological stress score	1.00	0.98 to 1.00	0.345

*Compared with bup/nal.
bup/nal, buprenorphine-naloxone; IVDU, intravenous drug use; MAP, Maudsley Addiction Profile.

Table 4 Logistic regression: predictors of continued non-prescribed opioid use by IVDU status

	No history of IVDU (n=1619)			History of IVDU (n=227)		
	Adjusted OR	95% CI	P value	Adjusted OR	95% CI	P value
Methadone use*	1.72	1.07 to 2.77	0.023	1.59	0.55 to 4.55	0.400
Female sex	0.78	0.56 to 1.10	0.156	1.01	0.56 to 1.85	0.962
Cannabis user	0.99	0.71 to 1.37	0.932	1.01	0.55 to 1.84	0.983
Currently working	1.28	0.91 to 1.82	0.159	1.23	0.56 to 2.69	0.604
Age	0.99	0.98 to 1.01	0.222	0.99	0.97 to 1.02	0.706
Married or common law	0.97	0.68 to 1.39	0.879	1.02	0.52 to 2.01	0.947
Non-prescribed benzodiazepine use	2.07	1.19 to 3.59	0.010	1.09	0.52 to 2.31	0.819
Opioid overdose history	1.28	0.90 to 1.82	0.176	1.35	0.76 to 2.40	0.311
MAP psychological stress score	0.99	0.97 to 1.01	0.342	1.01	0.98 to 1.05	0.406

*Compared with bup/nal.

bup/nal, buprenorphine-naloxone; IVDU, intravenous drug use; MAP, Maudsley Addiction Profile.

first-line recommendations by most organisational guidelines.^{5 6} Nonetheless, current evidence comparing MMT to bup/nal is mixed, and little is known about which patients will fare better on MMT compared with bup/nal.^{7 8} This is largely due to the limited applicability of the current evidence, such as due to the experimental trial designs, restrictive eligibility criteria and fixed-dosed schedule commonly employed by trials.⁷⁻⁹

Among a pragmatic sample of patients with OUD, the present study identified that there is no statistically significant association between OAT treatment type and response to treatment, as measured by ongoing non-prescribed opioid use, among patients receiving MMT or bup/nal over a 12-month period. However, we did find that patients who were on MMT were almost twice as likely to remain in treatment at the 12 months follow-up. We also found that among lower risk substance users, particularly those without a history of IVDU, MMT was associated with improved treatment response compared with bup/nal. This association was not seen among patients with a history of IVDU.

A Cochrane review published in 2014 showed that there was no difference between MMT and bup/nal when looking at non-prescribed substance use.⁸ They did, however, find that MMT was associated with better treatment retention compared with bup/nal.⁸ This is consistent with our findings. However, there is an emerging trend to shift away from treatment retention as an outcome for OUD treatments and rather focus on patient-centred measures.²⁹ A prior study of 2301 patients on OAT found that the majority of patients prioritised coming off OAT completely as the primary treatment goal, which is in direct contrast to the primary outcome of treatment retention used in most studies.¹⁰ While OAT has several reported benefits for OUD patients, little is known about the comparative effectiveness of these treatments on such outcomes in real clinical settings. A recent systematic review aimed to evaluate the comparative effectiveness on patient-centred outcomes including

sleep quality, global functioning and quality of life, but found that few to no studies examined these outcomes, making it difficult to draw any conclusions.³⁰ A more recent randomised controlled trial (RCT) analysing data from 272 participants found that patients on flexible dose bup/nal reported less cravings intensity and frequency compared with patients on MMT over 22 weeks of follow-up.³¹ Interestingly, they also found improved treatment retention among the MMT patients despite having more cravings.³¹

Given that MMT is a more potent opioid agonist without the ceiling effect associated with bup/nal, patients are more likely to experience a 'high' with methadone which they do not get with bup/nal.³² This may partly explain why MMT is associated with better treatment retention, without necessarily leading to improved treatment outcomes. More recent data are supporting rapid induction onto bup/nal, which preliminary evidence has associated with improved treatment retention.³³ Further data stratifying patients by induction method would be helpful to add to the discussion. Additionally, it is difficult to determine whether this benefit is solely due to the OAT or skewed due to cointervention bias secondary to the stringent treatment programmes and dispensing practices surrounding MMT use. Nonetheless, while methadone may have clear superiority over bup/nal when it comes to retention in treatment, further studies are required to evaluate more patient-centred outcomes.^{8 30 31}

We used ongoing non-prescribed use as an outcome, rather than focusing on relapse, as the purpose of OAT is harm reduction and patients do not necessarily consider complete abstinence to be the goal of intervention.^{34 35} Interestingly, we found that there was no difference between the two treatments, similar to prior studies. Given the higher potency of MMT at the mu-opioid receptor, it is sometimes preferred for patients with severe OUD, such as those with a history of IVDU, overdose or fentanyl use.^{12 36} However, while this is a general recommendation, it is not founded on evidence.

As such, we conducted a secondary analysis evaluating predictors of treatment response among patients with and without a history of IVDU, as a surrogate for severe OUD. IVDU is associated with the most rapid drug uptake and highest bioavailability compared with other routes of drug administration, thus causing people who use drugs to often have stronger cravings for the substance, as well as be at higher risk for complications from drug use (eg, infectious diseases, overdose).³⁷ We found that there was no difference in treatment response by treatment type among patients with a history of IVDU, but that among those without a history of IVDU, being on MMT was associated with a higher likelihood of ongoing non-prescribed opioid use. This contrasts with what is often recommended, as our study found that there was no difference among high-risk users, and bup/nal may be superior among low-risk users. This may be due to the fact that patients with an IVDU history are at higher risk for continued opioid use at baseline, and therefore, we are unable to detect a signal when comparing MMT to bup/nal. Given that the number of participants with a history of IVDU is also smaller, we may not be powered to detect this difference. That said, when we conducted the same analysis using overdose history as a surrogate for being a high-risk user, we once again found no difference in treatment response among those with a history of overdose ($n=161$), but that patients on bup/nal had superior treatment outcomes among those without a history of overdose ($n=708$, OR 0.55, 95% CI 0.32 to 0.94, $p=0.028$), as measured by ongoing non-prescribed opioid use. Further studies addressing this in a larger sample to allow for adequate subgroups, while controlling for co-interventions, would be valuable.

Strengths, limitations and future direction

A major strength of the study is that it included prospectively collected data that represents the current drug supply. This is in contrast to the majority of literature on OUD, which was conducted prior to the fentanyl crisis. Given that fentanyl is approximately 100 times more potent than morphine, there is increasing concern that prior dosing guidelines and treatment regimens may no longer be as effective in treating patients who use fentanyl, given its exponentially stronger potency.^{11 12} Our study is also strengthened by the large sample of patients and long follow-up period. We employed an objective primary outcome of ongoing non-prescribed opioid consumption, measured by UDSs, rather than relying on self-report. Furthermore, we imposed very minimal eligibility criteria, rendering the sample representative of the true patient population. A major strength of PSM is that it allows us to balance known confounders between the two treatment groups. While there may be other unmeasured or unknown confounders that cannot be accounted for, one would need to conduct an RCT to account for these confounders. If this RCT were to be conducted, it should follow a pragmatic trial design with flexible dosing schedules, large sample size, adequate

follow-up and realistic eligibility criteria. Moreover, the UDS assay we used is an immunoassay, which may be susceptible to false positives and negatives. However, given that we defined our primary outcome as having >50% of UDS positive for illicit opioids, and given that participants had their urines tested on a weekly to biweekly basis, we do not expect a few false positives or negatives to have an impact on our outcome. Further, we would expect these false results to be randomly distributed irrespective of treatment allocation, hence unlikely to bias the results.

Future studies should aim to further compare the effectiveness between MMT and bup/nal using patient-centred outcomes and pragmatic trial designs. The emphasis should be less on retention in treatment and more focused on substance use patterns, high-risk behaviours (eg, IVDU), quality of life measures and overdose risk.

CONCLUSIONS

Among a cohort of patients receiving OAT for OUD, we found that there is no statistically significant difference in ongoing non-prescribed opioid use between patients receiving MMT compared with bup/nal. Although we did find that patients on MMT are more likely to stay in treatment, it is unclear whether this correlates with improvements in patient-centred outcomes. We also found no association between treatment type and high-risk opioid consumption patterns among patients with a history of IVDU or opioid overdose, which does not support the recommendation to use MMT for patients with severe OUD, although severity of OUD can be broadly interpreted. This study adds to the current data on the comparative effectiveness of MMT and bup/nal, identifying no differences in ongoing opioid consumption even within a high-risk population. Further research is required to specifically investigate other important treatment outcomes including patient-centred outcomes, such as substance use patterns and quality of life measures, within a realistic sample of patients to help generate treatment recommendations that are precise and person centred.

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Patient consent for publication Consent obtained directly from patient(s).

Ethics approval This study involves human participants and was approved by Hamilton Integrated Research Ethics Board (#4556). Participants gave informed consent to participate in the study before taking part.

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Data availability statement Data are available on reasonable request.

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